



Nandaniya Sanjay Hajabhai
Staff Engineer

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8 Years of experience

PROFILE SUMMARY

Industry 4.0 and IIoT Architect with 8 years of expertise in designing Edge AI, Digital Twin, and cloud-native IIoT platforms across manufacturing, pharmaceutical, energy, and automotive industries. Proficient in backend development (Python, FastAPI, SQL), frontend technologies (React, TypeScript), and cloud services (AWS, Azure, GCP) to deliver scalable microservices, ETL pipelines, and asset management solutions. Experienced in containerized AI workload deployment on Qualcomm, NXP, and NVIDIA devices, OT—IT integration (OPC UA, MQTT, Kepware), and leading multi-plant digital transformations aligned with regulatory standards including 21 CFR Part 11 and IQ/OQ validation.

WORK EXPERIENCE

Staff Engineer

Nagarro

Mar 2024 - Present

- Collaborated with solution architects and customer engineering teams to deliver production-ready Industrial IoT and Digital Twin solutions supporting cloud modernization and enterprise integration.
- Built Data Engineering & ETL pipelines using AWS Lambda, AWS Glue, Amazon Redshift, and SQL to process, transform, and analyze high-volume industrial telemetry.
- Engineered scalable microservices and backend solutions with Python, SQL, TypeScript, FastAPI, Docker, and Kubernetes, optimizing API performance and enterprise application scalability.
- Delivered configurable multi-plant portfolio solutions enabling centralized asset management, production monitoring, KPI dashboards, historian integration, and alarm

KEY SKILLS

- Python
- FastAPI
- AWS
- Azure
- GCP
- Kubernetes
- Docker
- SQL
- React
- TypeScript
- Edge AI
- LLM
- RAG
- Computer Vision
- Kepware
- ML
- Artificial Intelligence
- Flask
- Full Stack
- MES
- IoT
- Integration
- Agentic Ai
- Javascript
- Ignition SCADA
- PTC Thingworx

CERTIFICATION

- AWS Certified Cloud Practitioner
- Microsoft Azure Fundamentals: AZ-900
- Thingworx
- PTC Kepware
- Ignition (Inductive Automation)
- McKinsey Forward Program

PROJECTS

Industrial Digital Twin & Smart Manufacturing Platform

6 Months

- Designed and delivered an enterprise Industrial Digital Twin platform for smart manufacturing.
- Developed ETL pipelines using AWS Glue, Lambda, Redshift, SQL, and Python.
- Built cloud-native microservices with FastAPI,

management.

- Designed Industrial IoT integrations using Ignition, ThingWorx, Kepware, MQTT, Sparkplug B, OPC UA, Modbus TCP/IP, BACnet, and REST APIs for seamless IT/OT connectivity.
- Architected hybrid cloud solutions on AWS, Azure, and GCP for industrial data ingestion, Digital Twin services, and scalable enterprise analytics.
- Developed responsive React (TypeScript) Portfolio and Plant Management applications with scalable Python/FastAPI backend APIs for asset hierarchy, Digital Twin visualization, and real-time monitoring.

Integration Engineer Specialist

Accenture

Apr 2023 - Feb 2024

- Delivered an enterprise-scale Azure Digital Twin enabling real-time asset visibility and faster operational decisions.
- Integrated live OT data from PLCs, SCADA, and smart devices using OPC frameworks and Azure IoT services.
- Built a scalable AKS-based ingestion and API layer supporting high-volume time-series data, modeling a standardized 11-level asset hierarchy from enterprise to machine level.
- Improved analytics performance through optimized KQL and Azure Data Explorer queries; centralized and stabilized ingestion using MS-SQL Datalogger and Azure services.
- Accelerated deployments and reliability through Azure DevOps CI/CD pipelines.
- Served as the primary client-facing technical lead, ensuring on-time delivery.

Control System Engineer

Wipro Technologies Ltd.

Nov 2021 - Apr 2023

- Implemented a cloud historian on Azure, consolidating time-series data across five sites, and enabled real-time monitoring and predictive insights using Azure Time Series Insights.
- Unified OT and IT data from GE Historian, Aspen, Kepware, and LIMS systems into a resilient ingestion architecture (Azure IoT Hub,

Docker, and Kubernetes and integrated industrial assets using OPC UA, MQTT Sparkplug B, Kepware, and Ignition.

Edge AI & Containerized AI Platform

8 Months

- Worked on end-to-end development and validation of containerized AI workloads for embedded edge platforms.
- Validated AI/ML containers for LLM, CV, and multimodal workloads across Qualcomm, NXP, and NVIDIA Jetson platforms ensuring performance and stability.
- Benchmarked LLM models, including quantization, latency, and throughput, and designed FAISS-based validation workflows.

IoT Solutions & Structures

15 Months

- Designed AWS-based Industrial IoT platforms for edge-to-cloud connectivity across pharma, energy, automotive, and building materials domains.
- Developed scalable microservices and integrated industrial assets using OPC UA, BACnet, MQTT, Kepware, and Ignition.
- Implemented AWS SiteWise, TwinMaker, and cloud automation services to enable real-time monitoring and analytics.

Digital Twin On Microsoft Azure

12 Months

- Led the development and implementation of a cutting-edge Digital Twin project on Microsoft Azure.
- Created virtual representations of physical assets, integrated real-time data streams, and leveraged Azure services for connectivity and analytics.

Azure Digital Twin Implementation

6 Months

- Implemented an enterprise Azure Digital Twin solution by integrating PLCs, SCADA, OPC servers, and industrial devices.
- Developed Python APIs and optimized KQL queries for high-performance analytics.

Automatic Water Distribution System and Theft Control in Smart city

6 Months

Stream Analytics, Data Lake).

- Increased platform adoption through custom dashboards and notification frameworks.
- Standardized IIoT connectivity across PLCs, SCADA, and metering systems, improving data quality by refactoring and normalizing JSON schemas.
- Led sprint planning and delivery while maintaining daily client alignment.

Executive Engineer

Torrent Pharmaceutical Ltd.

Jun 2019 - Oct 2021

- Led end-to-end MES—L2—L1 integration, integrating 100+ machines via Kepware OPC UA, centralized SCADA/historian, Syncade MES, and SQL-based reporting, with workflow and recipe design improving production control.
- **Delivered 21 CFR Part 11—compliant upgrades** across PLCs, HMIs, SCADAs, and IQMS systems — audit trails, user access controls, SSRS web reporting, networking, and IQ/OQ validation of new and existing equipment.

Graduate Trainee

Sanofi Ltd.

Sep 2018 - Jun 2019

- Supported instrumentation and automation activities in pharmaceutical manufacturing.
- Assisted in preventive maintenance, calibration, commissioning, and troubleshooting of process instruments.
- Worked with PLCs, HMIs, SCADA systems, transmitters, control valves, and industrial sensors.
- Supported equipment qualification, documentation, and compliance activities.

INTERNSHIP

Government Engineering College, Rajkot
Automatic Water Distribution System and Theft Control in Smart city

6 Months

- Developed an automatic water distribution system aimed at precise flow allocation and reduction of water wastage.

- Developed an automatic water distribution system with Arduino and GSM module integrations to manage flow allocation and theft detection.

LANGUAGES

English

Gujarati

Hindi

★ HOBBIES

Technical Blogging

Mentoring

Open Source Contribution

★ ACHIEVEMENTS & RECOGNITION

Recognition

Received Nagarro PM & Lead Appreciation and Nagarro Care Award (2×) for Edge AI validation and delivery excellence.

★ EXTRA-CURRICULAR ACTIVITY

Technical Workshops & Training

Conducted and participated in technical workshops related to IIoT, Edge AI validation, and industrial automation tools.

SOCIAL LINKS

www.linkedin.com/in/sanjay-nandaniya-1585b4142

- Integrated theft control mechanism using Arduino and GSM modules for municipal corporations and residential areas ensuring efficient resource management.

EDUCATION

B.Tech / B.E. - Instrumentation and Control Engineering

2018

Government Engineering College, Rajkot

Grade - 7.9/10

Class XII

2014

Gujarat, Gujarati

Marks - 76%

Class X

2012

Gujarat, Gujarati

Marks - 77%